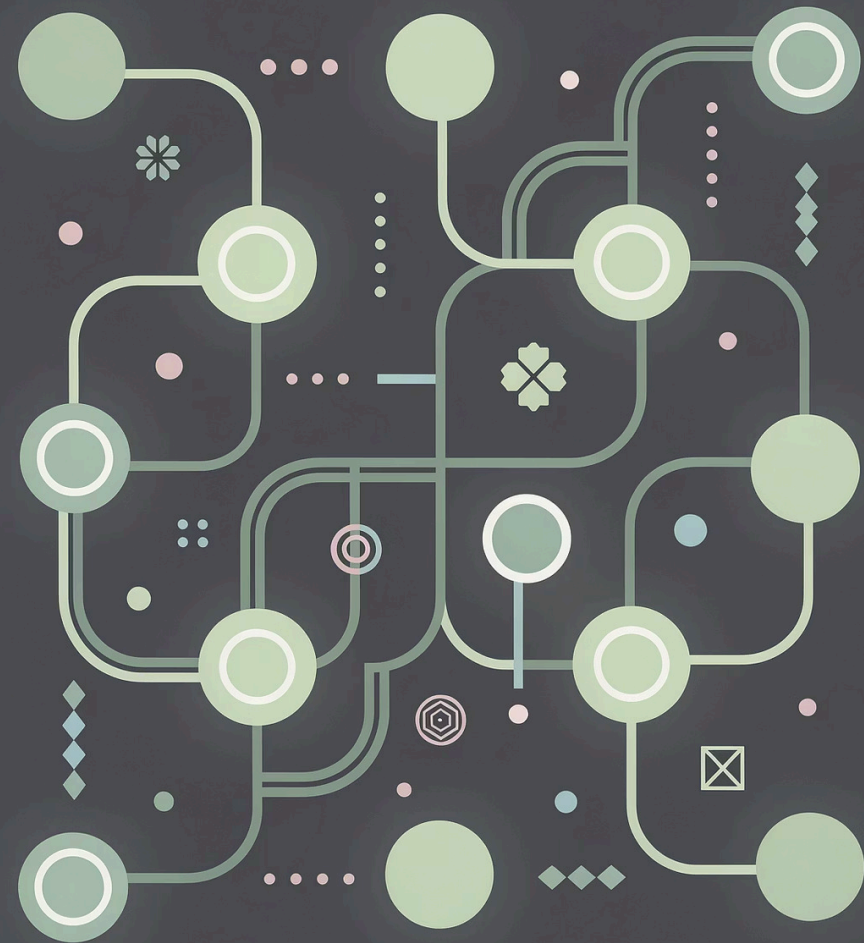


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Inside the Engine: Attention & the GPT Architecture

A deep dive into the two foundational innovations that power today's most capable large language models — the Attention Mechanism and the Transformer decoder stack.



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CHAPTER 1

The Attention Mechanism

How modern language models learned to focus on what matters most — all at once.



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The Problem: Sequential Processing

Before attention, models like RNNs and LSTMs processed text **sequentially** — one token at a time, left to right, much like reading a sentence word by word. Each word's representation depended heavily on the hidden state passed from the previous step.

Information Bottleneck

Long-range dependencies were compressed into a single fixed-size vector, causing critical context to be lost over long sequences.

No Parallelism

Sequential computation meant training was painfully slow — each step had to wait for the previous one to finish, preventing efficient GPU utilization.

Vanishing Gradients

Signals degraded across many time steps, making it difficult for models to learn relationships between distant words in a sentence.

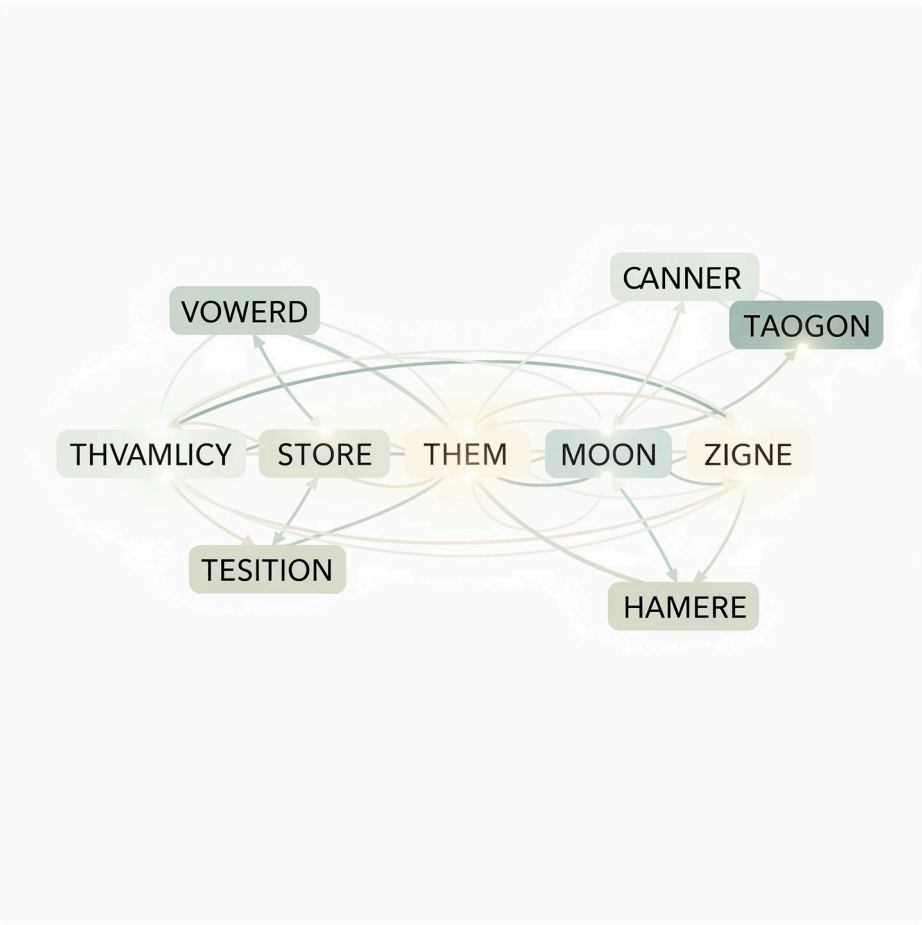
The Breakthrough: Attention

The attention mechanism revolutionized NLP by allowing a model to **simultaneously weigh the importance of every word** in an input sequence when processing each individual word. Instead of compressing everything into a single bottleneck vector, the model can dynamically attend to the most relevant parts of the input — no matter how far apart they are.

A Classic Example

| *"The animal didn't cross the street because **it** was too tired."*

What does **"it"** refer to? A human instantly knows it means "the animal." The attention mechanism learns this too — by assigning a high attention weight between "it" and "The animal," even though they are several tokens apart. This contextual awareness is what makes modern LLMs remarkably powerful.



How Attention Works: Q, K, V

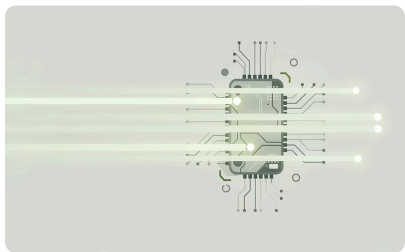
At its core, attention computes a weighted sum of **Values** based on the compatibility between **Queries** and **Keys**. Each input token is projected into three learned vectors, and their interactions determine where the model "looks."



This mechanism — known as **Scaled Dot-Product Attention** — runs in parallel across all positions, enabling massive speedups over recurrent approaches. Multi-head attention extends this further by running several attention operations simultaneously, each learning different types of relationships (syntactic, semantic, positional).

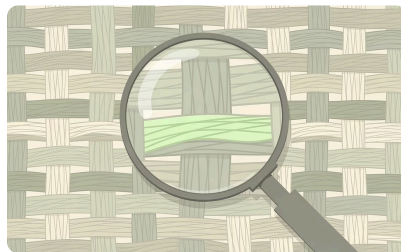
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Why Attention Changes Everything



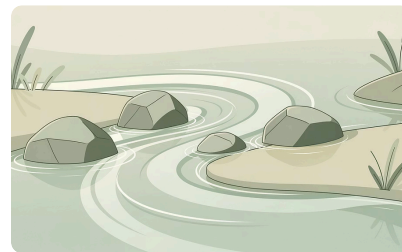
Parallel Processing

Every token attends to every other token simultaneously, unlocking massive GPU parallelism and dramatically reducing training time compared to sequential RNNs.



Long-Range Context

No more vanishing gradients over distance. Attention connects any two positions in a sequence directly, enabling models to capture dependencies across thousands of tokens.



Dynamic Weighting

Attention weights are input-dependent — the model adapts its focus based on what it's reading, not a fixed pattern. This makes representations richly contextual.



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CHAPTER 2

The Architecture of GPT

From the original Transformer to the decoder-only design that powers the world's most capable generative models.

The Transformer: A New Blueprint

Introduced in the landmark 2017 paper *"Attention Is All You Need,"* the **Transformer architecture** replaced recurrence entirely with self-attention. The original design featured two major components working in tandem:

Encoder

Reads and compresses the entire input sequence into a rich set of contextual representations. Used in models like BERT for understanding tasks — classification, named entity recognition, and semantic similarity.

Decoder

Takes the encoder's representations (or its own previous outputs) and generates text one token at a time, autoregressively. Uses **masked self-attention** to prevent "peeking" at future tokens during generation.

- ❏ Models like GPT use **only the decoder** stack. Models like BERT use **only the encoder**. The full encoder-decoder design is used in models like T5 and the original machine translation Transformers.

GPT's Decoder-Only Design

The GPT series — **Generative Pre-trained Transformer** — strips the architecture down to the decoder stack. This design choice is deliberate: generative tasks like next-token prediction only require the model to attend to **previous** context, never future tokens. Here's what happens inside a single decoder block:

01

Masked Multi-Head Self-Attention

Each token attends to all preceding tokens (but not future ones). Multiple attention heads capture different relationship types — syntactic structure, semantic meaning, and positional patterns — in parallel.

03

Feed-Forward Network

A two-layer MLP with a nonlinear activation (typically GELU) processes each position independently, expanding and then compressing the representation to capture complex features.

02

Add & Normalize

Residual connections add the attention output back to the input, preserving the original signal. Layer normalization then stabilizes values, enabling deeper stacking without degradation.

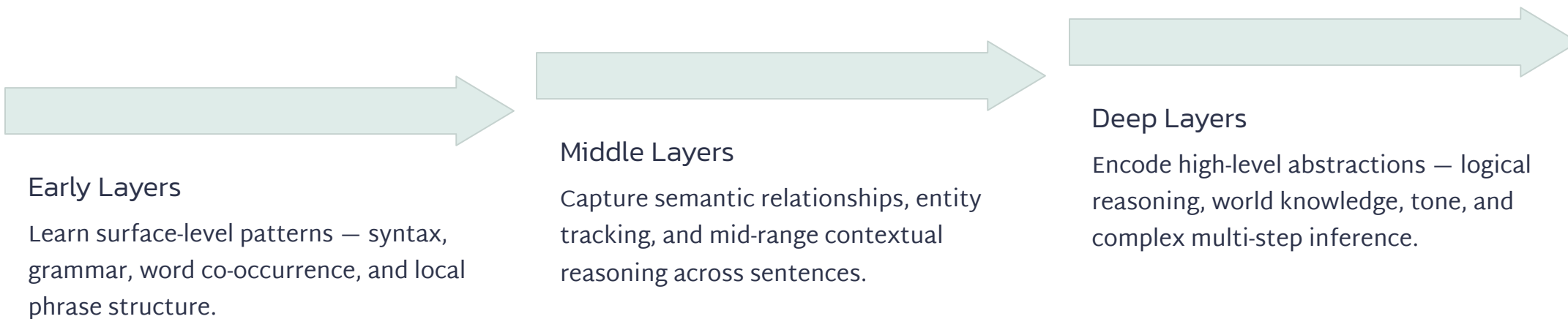
04

Add & Normalize (Again)

Another residual connection and normalization step. The output of this block becomes the input to the next decoder block in the stack.

The Power of Depth: Stacking Decoder Blocks

A single decoder block captures relatively simple patterns. The true power of GPT comes from **stacking dozens — even hundreds — of these blocks** on top of each other. Each successive layer builds increasingly abstract and nuanced representations of language.



GPT-3 uses **96 decoder blocks** with 175 billion parameters. GPT-4 is believed to scale even further. This depth is what enables coherent, contextually rich text generation over long passages.

Scale at a Glance

The evolution of GPT models demonstrates how scaling depth, width, and data drives dramatic capability improvements.

Model	Parameters	Decoder Layers	Training Data
GPT-1	117M	12	~5 GB text
GPT-2	1.5B	48	~40 GB text
GPT-3	175B	96	~570 GB text
GPT-4	Undisclosed	Undisclosed	Massive multi-modal

Each generation leap brought not just quantitative improvement, but qualitative new capabilities — from basic text completion to multi-step reasoning, code generation, and creative writing.

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Putting It All Together

The attention mechanism and the decoder-only Transformer architecture form a powerful combination that defines the modern LLM paradigm.

Attention

Enables parallel, context-aware processing of every token in a sequence — the computational engine.

Decoder Stack

Layers attention blocks into a deep generative pipeline optimized for next-token prediction.

Scale

Massive parameters and data transform simple architectural principles into emergent intelligence.

100%

Attention-Based
No recurrence needed

96

Decoder Layers
In GPT-3's architecture

175B

Parameters
GPT-3's learned weights

Key Takeaways

1 Attention replaces recurrence

By weighing all tokens simultaneously, the attention mechanism eliminated sequential bottlenecks and enabled models to capture long-range dependencies directly.

2 Decoder-only is purpose-built for generation

GPT's architecture discards the encoder in favor of a masked decoder stack, perfectly suited for autoregressive text generation tasks.

3 Depth unlocks emergent capabilities

Stacking decoder blocks creates hierarchical representations — from syntax in early layers to abstract reasoning in deep layers — that scale with parameters and data.

